Endogenizing Nature in Production Theory: A Framework of Economic Commensalism

Fernando Souza da Vieira[1] Alexandre Chibebe Nicolella[2]

[1]Faculty of Economics, Administration and Accounting of Ribeirão Preto, University of São Paulo, Brazil

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer nec odio. Praesent libero. Sed cursus ante dapibus diam. Sed nisi. Nulla quis sem at nibh elementum imperdiet. Duis sagittis ipsum. Praesent mauris. Fusce nec tellus sed augue semper porta. Mauris massa. Vestibulum lacinia arcu eget nulla.

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Curabitur sodales ligula in libero. Sed dignissim lacinia nunc. Curabitur tortor. Pellentesque nibh. Aenean quam. In scelerisque sem at dolor. Maecenas mattis. Sed convallis tristique sem. Proin ut ligula vel nunc egestas porttitor.

Nullam quis risus eget urna mollis ornare vel eu leo. Aenean lacinia bibendum nulla sed consectetur. Vestibulum id ligula porta felis euismod semper.

Keywords: economic commensalism; environmental economics; rational incentives; commons; carbon markets.

1. Introduction

From an intertemporal social-efficiency perspective, neither extreme predation, defined by the indiscriminate and often illegal depletion of natural resources, nor strict preservationism, characterized by minimal or non-extractive use of the environment, can provide a stable and efficient resolution of the trade-off between production and ecological integrity (DASGUPTA, 2001). These two poles represent opposing boundaries of environmental interaction: the former disregards sustainability entirely, while the latter may overly constrain productive potential under conventional productivity metrics, which typically emphasize output expansion over ecological feedbacks. Between these extremes lies the economic problem of resource extraction: the allocation of renewable and non-renewable

[1] fernandodvieira@usp.br
[2] nicolella@usp.br

natural resources over time, under conditions of scarcity, regeneration, depletion, opportunity costs, and intertemporal trade-offs (HOTELLING, 1931; CLARK, 2010; DASGUPTA, 2001). In this narrower economic sense, extraction need not be predatory; it may remain compatible with sustainability when the rate of use respects regenerative capacity, stock constraints, and the shadow value of future resource availability.

The concept of extractivism, however, carries a different meaning in the political economy literature. Rather than denoting resource extraction as an intertemporal allocation problem, extractivism refers to a development pattern based on the large-scale appropriation of natural resources, frequently oriented toward commodity exports, limited domestic value addition, and ecological degradation (GUDYNAS et al., 2009; ACOSTA, 2013; MCKAY, 2017). Under this interpretation, the issue is not extraction per se, but the institutional and productive regime through which extraction becomes structurally tied to depletion, dependency, and asymmetric distribution of environmental costs.

–> Mostrar que a economia tradicional não ignora completamente a natureza, mas tende a tratá-la como estoque, restrição, externalidade, renda da terra ou custo ambiental;

–> Dizer que o problema do artigo é outro: a natureza raramente aparece como fator produtivo endógeno, dinâmico e estrategicamente incorporado à decisão econômica;

–> Introduzir, com cautela, a lacuna que justificará o economic commensalism. This distinction is central to the argument developed here. The paper does not treat extraction and extractivism as interchangeable terms. Instead, it uses the gap between them to identify the conditions under which resource use moves from an economically rational allocation problem to an ecologically predatory regime, as reflected in declining future productivity, ecological threshold violations, divergence between private and social payoffs, and intertemporal welfare losses.

2. The Limits of Current Environmental Governance

—> Mecanismos atuais de governança ambiental (regulação, precificação, mercados de carbono, etc.) são insuficientes para lidar com a complexidade e a dinâmica dos sistemas socioecológicos.

—> Dados e evidências mostram que, apesar dos esforços, as tendências de degradação ambiental e perda de biodiversidade continuam, indicando falhas estruturais nos modelos de governança atuais.

—> Citar esforços internacionais (Acordo de Paris, Objetivos de Desenvolvimento Sustentável) e nacionais (políticas ambientais, leis de proteção) que não têm conseguido reverter as tendências negativas.

—> Usar dados de desmatamento, perda de biodiversidade, emissões de gases de efeito estufa, Brasil e mundo, para ilustrar a insuficiência dos mecanismos atuais.

–> Concluir que é necessário repensar os fundamentos teóricos e práticos da governança ambiental, buscando abordagens que integrem melhor a natureza como parte dos sistemas econômicos e sociais, e que sejam mais adaptativas, inclusivas e eficazes.

3. Literature Review: From Externalities to Incentives

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

4. Model

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

5. Results

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

6. Discussion

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

7. Final Remarks

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin vitae ligula tortor. Suspendisse erat orci, fermentum in consectetur id, hendrerit nec erat. Donec ac gravida urna. Fusce a est sed ligula congue posuere. Suspendisse dictum erat ut velit fringilla pellentesque. Pellentesque ante sem, accumsan tincidunt diam vitae, convallis aliquam dui. Nulla nec nulla nec dui ultrices tempor. Vivamus sit amet facilisis dolor, nec luctus tortor. Fusce tempor eget eros a bibendum. Donec sed dui vel dolor tempor iaculis. Duis laoreet risus nisl, eu maximus erat pellentesque sit amet. Morbi sodales metus sit amet lectus dictum mollis. Vestibulum nunc neque, tempus eu posuere eget, malesuada et massa. Nullam ullamcorper consequat diam quis cursus. Phasellus ultricies sollicitudin eros quis dictum. Duis molestie venenatis arcu eget fermentum.

Suspendisse condimentum sed nibh et bibendum. Etiam a placerat tellus. In id tellus purus. Pellentesque sagittis porttitor augue, nec commodo tellus facilisis eget. Phasellus mattis dictum pharetra. Nulla euismod, arcu ut pretium posuere, quam magna scelerisque ligula, ut blandit sem sem in eros. Sed rhoncus turpis vitae ex porttitor, a posuere velit aliquet. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Quisque at ligula molestie, ornare dui vitae, placerat lectus.

Praesent porttitor neque quis finibus vulputate. Suspendisse potenti. Curabitur eu turpis ac elit dignissim venenatis. Nam porttitor ligula ac sollicitudin gravida. Donec leo magna, dapibus id finibus id, suscipit vitae lacus.

Suspendisse sit amet libero ac enim tempus cursus quis quis nisl. Nullam a risus vel augue eleifend iaculis. Proin vel posuere enim, accumsan congue massa. Nam venenatis, nibh non cursus auctor, quam ex congue tellus, in dictum arcu justo quis risus. Etiam sollicitudin, turpis ut convallis semper, leo dui elementum lectus, sit amet faucibus arcu ligula id quam. Cras nec justo sit amet risus mattis convallis. Duis cursus et urna quis malesuada. Maecenas facilisis porttitor felis, a ultrices dolor sodales vitae. Morbi eu consequat ante. Maecenas suscipit nec urna at finibus. Maecenas interdum rutrum mauris, sit amet viverra mauris finibus in. In mollis semper mauris, dictum tincidunt libero bibendum nec. Praesent sed felis non massa commodo molestie. Cras gravida odio et ante auctor auctor.

Morbi quis lacus fermentum, malesuada lectus at, bibendum nunc. Morbi mi odio, rutrum sed euismod vitae, tempus eget justo. Nullam faucibus lacus et sodales rutrum. Aliquam finibus augue id velit euismod, at maximus eros ullamcorper. In hac habitasse platea dictumst. Aenean vel maximus nunc. Nunc in ultricies ex. Mauris metus tortor, tempor nec tortor a, pretium hendrerit risus. Nulla sed porta risus, sed lobortis est. Nam lobortis velit porta feugiat ultrices. Phasellus sodales mauris et magna convallis sodales. Proin tincidunt libero et lacus posuere porta. Vestibulum tellus ante, condimentum a diam vel, eleifend pellentesque lacus.

Sed scelerisque eros massa, id congue justo faucibus ac. Donec dolor nisi, faucibus euismod erat in, dictum iaculis mi. Duis nec venenatis mauris. Proin tincidunt nisl nec maximus egestas. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur fringilla luctus nulla, nec faucibus sem pharetra sed. Ut venenatis odio eget fermentum aliquam. Sed ac ex velit. Vestibulum non euismod libero. Sed tincidunt turpis odio, id aliquet lorem pretium vel.

References

ACOSTA, A. Extractivism and neoextractivism: two sides of the same curse. **Beyond development: alternative visions from Latin America**, Quito, v. 1, p. 61–86, 2013.

CLARK, C. W. **Mathematical bioeconomics: the mathematics of conservation**. [S.l.]: John Wiley & Sons, 2010.

DASGUPTA, P. **Human Well-Being and the Natural Environment**. 1st. ed. Oxford: Oxford University Press, 2001. ISBN 9780199247882.

GUDYNAS, E. et al. Diez tesis urgentes sobre el nuevo extractivismo. **Extractivismo, política y sociedad**, Quito, Ecuador, v. 187, p. 187–225, 2009.

HOTELLING, H. The economics of exhaustible resources. **Journal of political Economy**, The University of Chicago Press, v. 39, n. 2, p. 137–175, 1931.

MCKAY, B. M. Agrarian extractivism in bolivia. **World Development**, Elsevier, v. 97, p. 199–211, 2017.